

ScienceQtech Employee Performance Mapping –

DESCRIPTION –

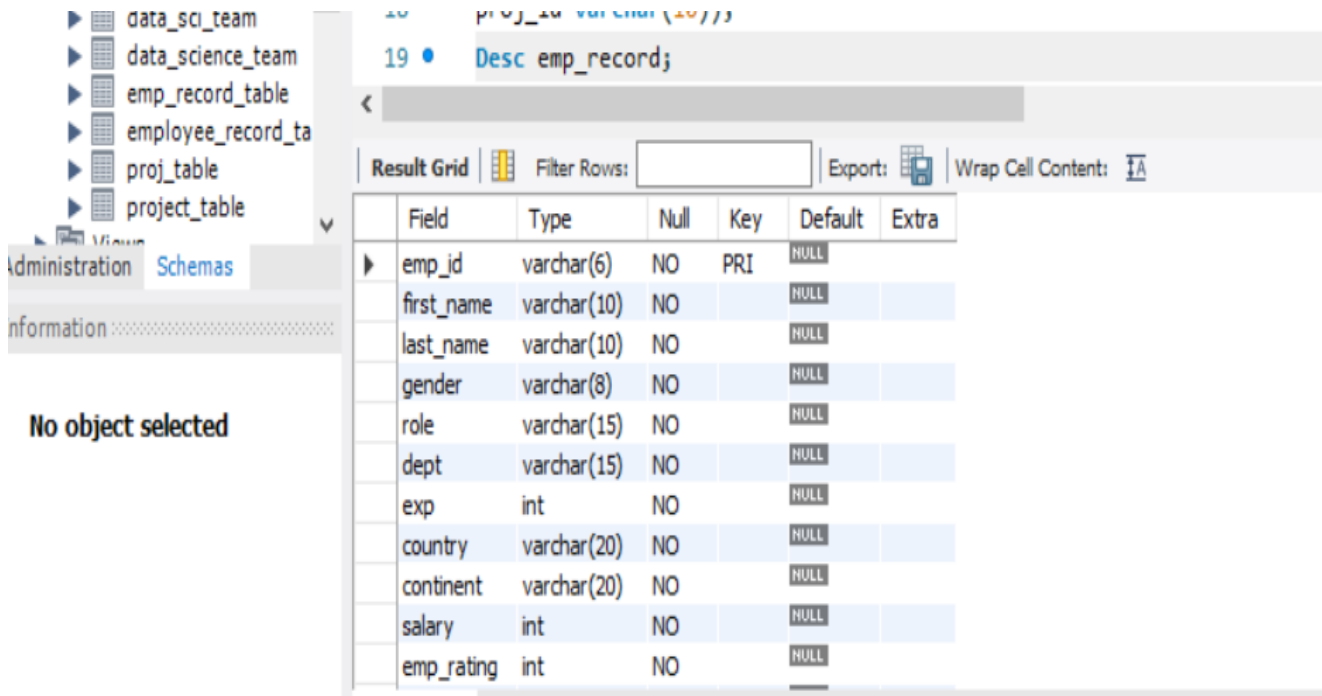
ScienceQtech is a startup that works in the Data Science field. ScienceQtech has worked on fraud detection, market basket, self-driving cars, supply chain, algorithmic early detection of lung cancer, customer sentiment, and the drug discovery field. With the annual appraisal cycle around the corner, the HR department has asked you (Junior Database Administrator) to generate reports on employee details, their performance, and on the project that the employees have undertaken, to analyze the employee database and extract specific data based on different requirements. Objective: To facilitate a better understanding, managers have provided ratings for each employee which will help the HR department to finalize the employee performance mapping. As a DBA, you should find the maximum salary of the employees and ensure that all jobs are meeting the organization's profile standard. You also need to calculate bonuses to find extra cost for expenses. This will raise the overall performance of the organization by ensuring that all required employees receive training. Note: You must download the dataset from the course resource section in LMS and create a table to perform the above objective.

Dataset description –

```
Create database employee;
Use employee;
Create table emp_record
(emp_id varchar (6) not null primary key,
first_name varchar (10) not null,
last_name varchar (10) not null,
gender varchar (8) not null,
role varchar (15) not null,
dept varchar (15) not null,
exp int not null,
country varchar (20) not null,
continent varchar (20) not null,
salary int not null,
emp_rating int not null,
manager_id varchar (10),
proj_id varchar (10));
```

Output -

Desc table emp_record;



Administration Schemas

No object selected

Result Grid

Field	Type	Null	Key	Default	Extra
emp_id	varchar(6)	NO	PRI	NULL	
first_name	varchar(10)	NO		NULL	
last_name	varchar(10)	NO		NULL	
gender	varchar(8)	NO		NULL	
role	varchar(15)	NO		NULL	
dept	varchar(15)	NO		NULL	
exp	int	NO		NULL	
country	varchar(20)	NO		NULL	
continent	varchar(20)	NO		NULL	
salary	int	NO		NULL	
emp_rating	int	NO		NULL	

SQL Code –

```
Create table pro_table (  
project_id varchar (10) not null primary key,  
proj_name varchar (20) not null,  
domain varchar (10) not null,  
start_date date not null,  
closure_date date not null,  
dev_qtr varchar (5) not null,  
status varchar (10) not null);  
Desc table pro_table;
```

Output –

SCHEMAS

Filter objects

- ecommercedb
 - employee**
 - Tables
 - data_sci_team
 - data_science_team
 - emp_record_table
 - employee_record_ta
 - proj_table
 - project_table

Administration Schemas

Information

No object selected

```

23 • create table proj_table
24   (project_id varchar(10) not null primary key,
25    proj_name varchar(20) not null,
26    domain varchar(10) not null,
27    start_date date not null,
28    closure_date date not null,
29    dev_qtr varchar(5) not null,
30    status varchar(10) not null);
31 • desc proj_table;

```

Limit to 1000 rows

Result Grid

Filter Rows:

Export: Wrap Cell Content:

	Field	Type	Null	Key	Default	Extra
▶	project_id	varchar(10)	NO	PRI	NULL	
	proj_name	varchar(20)	NO		NULL	
	domain	varchar(10)	NO		NULL	
	start_date	date	NO		NULL	
	closure_date	date	NO		NULL	
	dev_qtr	varchar(5)	NO		NULL	

Result 4 x

SQL Code –

```

Create table data_sci_team (
  emp_id varchar (10) not null primary key,
  f_name varchar (10) not null,
  l_name varchar (10) not null,
  gender varchar (8) not null,
  role varchar (20) not null,
  dept varchar (20) not null,
  exp int not null,
  country varchar (15) not null,
  continent varchar (200) not null);
desc data_sci_team;

```

Output -

SCHEMAS

Filter objects

- ecommercedb
 - employee
 - Tables
 - data_sci_team
 - data_science_team
 - emp_record_table
 - employee_record_ta
 - proj_table
 - project_table

Administration Schemas

Information

No object selected

```

35 • create table data_sci_team
36   (emp_id varchar(10) not null primary key,
37    f_name varchar(10) not null,
38    l_name varchar(10) not null,
39    gender varchar(8) not null,
40    role varchar(20) not null,
41    dept varchar(20) not null,
42    exp int not null,
43    country varchar(15) not null,
44    continent varchar(200) not null);
  
```

Result Grid

Field	Type	Null	Key	Default	Extra
emp_id	varchar(10)	NO	PRI	NULL	
f_name	varchar(10)	NO		NULL	
l_name	varchar(10)	NO		NULL	
gender	varchar(8)	NO		NULL	
role	varchar(20)	NO		NULL	

Result 7 x

Output

- Import data into tables.
SQL code: Imported using import function on workbench.
Output: emp_record_table
Select * from emp_record_table;

20 • `select * from emp_record_table;`
21

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

EMP_ID	FIRST_NAME	LAST_NAME	GENDER	ROLE	DEPT	EXP	COUNTRY	CONTINENT	SALARY	EMP_RATING
E001	Arthur	Black	M	PRESIDENT	ALL	20	USA	NORTH AMERICA	16500	5
E005	Eric	Hoffman	M	LEAD DATA SCIENTIST	FINANCE	11	USA	NORTH AMERICA	8500	3
E010	William	Butler	M	LEAD DATA SCIENTIST	AUTOMOTIVE	12	FRANCE	EUROPE	9000	2
E052	Dianna	Wilson	F	SENIOR DATA SCIENTIST	HEALTHCARE	6	CANADA	NORTH AMERICA	5500	5
E057	Dorothy	Wilson	F	SENIOR DATA SCIENTIST	HEALTHCARE	9	USA	NORTH AMERICA	7700	1
E083	Patrick	Voltz	M	MANAGER	HEALTHCARE	15	USA	NORTH AMERICA	9500	5
E103	Emily	Grove	F	MANAGER	FINANCE	14	CANADA	NORTH AMERICA	10500	4
E204	Karene	Nowak	F	SENIOR DATA SCIENTIST	AUTOMOTIVE	8	GERMANY	EUROPE	7500	5
E245	Nian	Zhen	M	SENIOR DATA SCIENTIST	RETAIL	6	CHINA	ASIA	6500	2
E260	Roy	Collins	M	SENIOR DATA SCIENTIST	RETAIL	7	INDIA	ASIA	7000	3

emp_record_table 8 x | Read Only | Conte

Output

Select * from project_table;

31 • `select * from project_table;`
32 • `select * from project_table;`
33

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

PROJECT_ID	PROJ_NAME	DOMAIN	START_DATE	CLOSURE_DATE	DEV_QTR	STATUS
P103	Drug Discovery	HEALTHCARE	04-06-2021	6/20/2021	Q1	DONE
P105	Fraud Detection	FINANCE	04-11-2021	6/25/2021	Q1	DONE
P109	Market Basket Analysis	RETAIL	04-12-2021	6/30/2021	Q1	DELAYED
P204	Supply Chain Management	AUTOMOTIVE	07/15/2021	9/28/2021	Q2	WIP
P302	Early Detection of Lung Cancer	HEALTHCARE	10-08-2021	12/18/2021	Q3	YTS
P406	Customer Sentiment Analysis	RETAIL	07-09-2021	9/24/2021	Q2	WIP

Select * from data_science_team;

ecommerceup

employee

- Tables
 - data_sci_team
 - data_science_team
 - emp_record_table
 - employee_record_ta
 - proj_table
 - project_table

Administration Schemas

Information

No object selected

```

45 • desc data_sci_team;
46 • select * from data_science_team;
47 • select * from emp_record_table;
48 • select emp_id, first_name, last_name, gender, dept from emp_record_table;

```

Result Grid

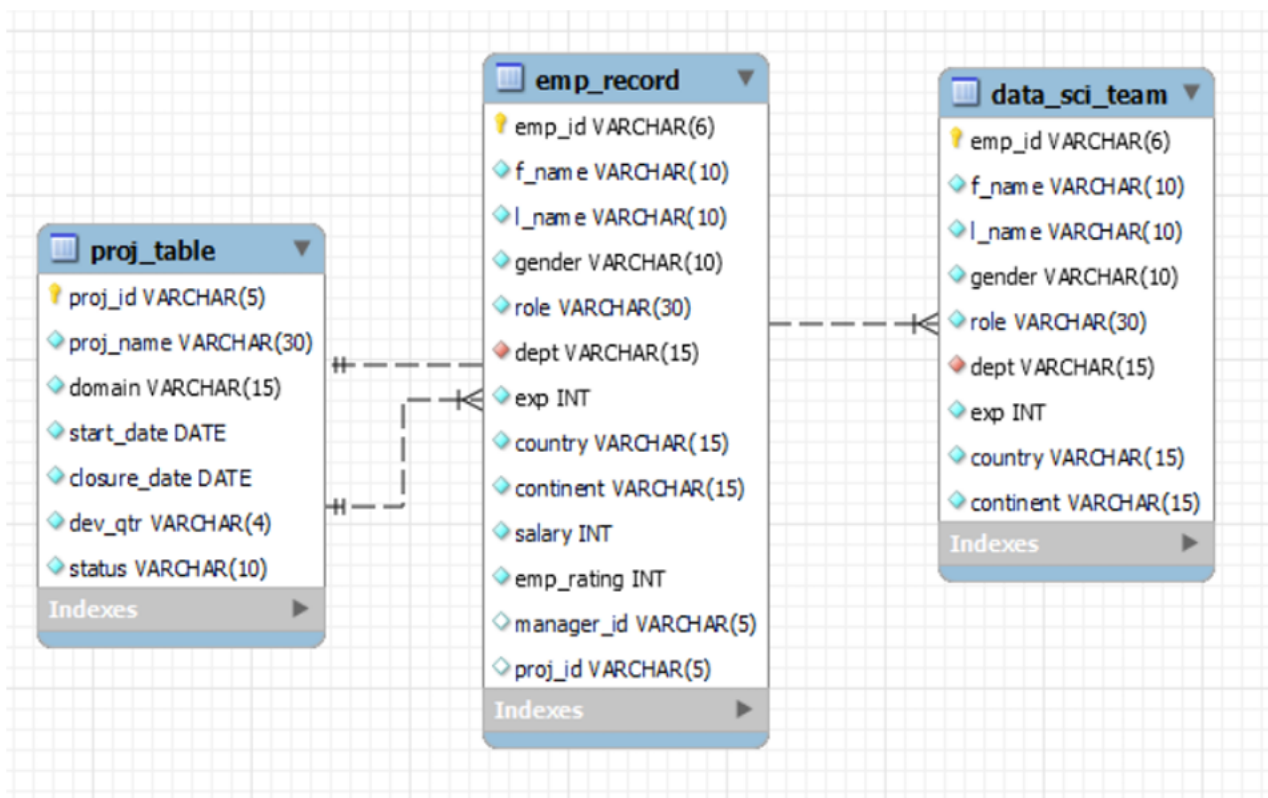
	EMP_ID	FIRST_NAME	LAST_NAME	GENDER	ROLE	DEPT	EXP	COUNTRY	CONTINENT
▶	E005	Eric	Hoffman	M	LEAD DATA SCIENTIST	FINANCE	11	USA	NORTH AMERICA
	E010	William	Butler	M	LEAD DATA SCIENTIST	AUTOMOTIVE	12	FRANCE	EUROPE
	E052	Dianna	Wilson	F	SENIOR DATA SCIENTIST	HEALTHCARE	6	CANADA	NORTH AMERICA
	E057	Dorothy	Wilson	F	SENIOR DATA SCIENTIST	HEALTHCARE	9	USA	NORTH AMERICA
	E204	Karene	Nowak	F	SENIOR DATA SCIENTIST	AUTOMOTIVE	8	GERMANY	EUROPE
	E245	Nian	Zhen	M	SENIOR DATA SCIENTIST	RETAIL	6	CHINA	ASIA
	E260	Roy	Collins	M	SENIOR DATA SCIENTIST	RETAIL	7	INDIA	ASIA
	E403	Steve	Hoffman	M	ASSOCIATE DATA SCIENTIST	FINANCE	4	USA	NORTH AMERICA
	E478	David	Smith	M	ASSOCIATE DATA SCIENTIST	RETAIL	3	COLOMBIA	SOUTH AMERICA
	E505	Chad	Wilson	M	ASSOCIATE DATA SCIENTIST	HEALTHCARE	5	CANADA	NORTH AMERICA
	E532	Claire	Brennan	F	ASSOCIATE DATA SCIENTIST	AUTOMOTIVE	3	GERMANY	EUROPE

data science team 10 x

Create an ER diagram from employee database

SQL code: Reverse Engineering of employee database.

Output:

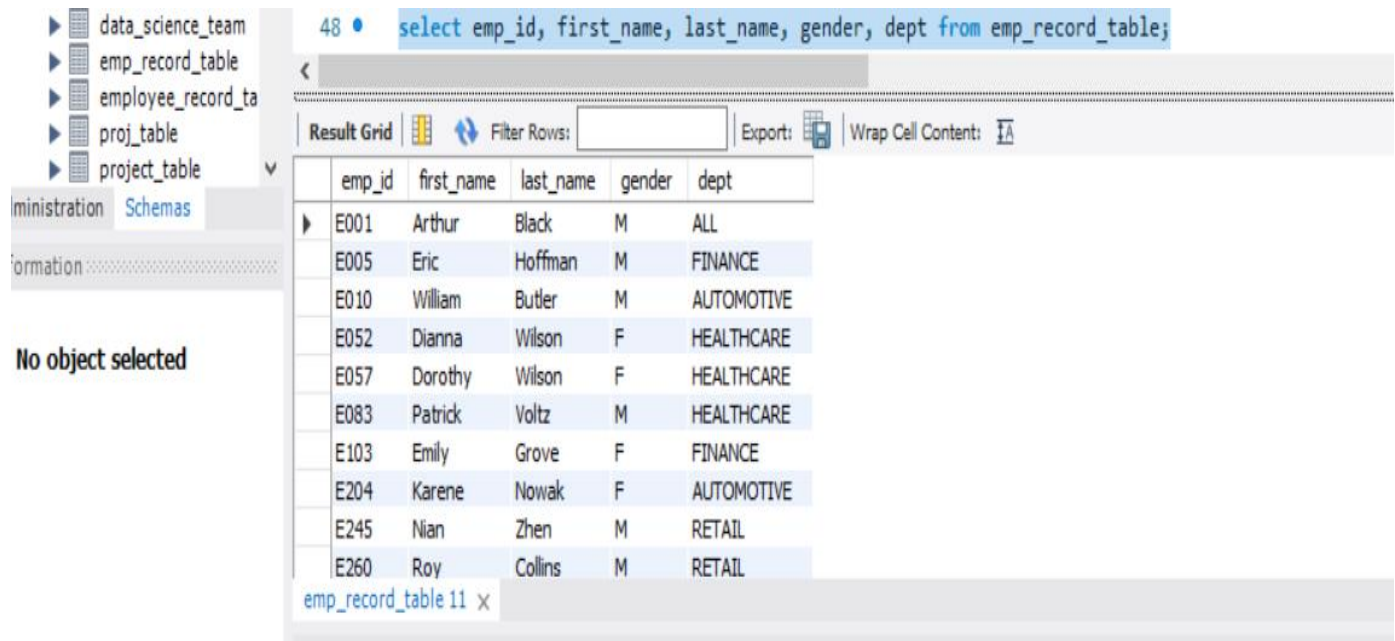


Write a query to fetch EMP_ID, FIRST_NAME, LAST_NAME, GENDER, and DEPARTMENT from the employee record table, and make a list of employees and details of their department.

SQL code –

```
select emp_id, first_name, last_name, gender, dept from emp_record_table;
```

Output –



The screenshot shows a database management system interface. On the left, a tree view lists several tables: data_science_team, emp_record_table, employee_record_table, proj_table, and project_table. The 'Schemas' tab is selected. Below the tree, it says 'No object selected'. In the center, a SQL query is entered in a text box: 'select emp_id, first_name, last_name, gender, dept from emp_record_table;'. Below the query, a 'Result Grid' is displayed, showing a table with 6 columns: emp_id, first_name, last_name, gender, and dept. The table contains 10 rows of data. At the bottom of the result grid, there is a tab labeled 'emp_record_table 11'.

emp_id	first_name	last_name	gender	dept
E001	Arthur	Black	M	ALL
E005	Eric	Hoffman	M	FINANCE
E010	William	Butler	M	AUTOMOTIVE
E052	Dianna	Wilson	F	HEALTHCARE
E057	Dorothy	Wilson	F	HEALTHCARE
E083	Patrick	Voltz	M	HEALTHCARE
E103	Emily	Grove	F	FINANCE
E204	Karene	Nowak	F	AUTOMOTIVE
E245	Nian	Zhen	M	RETAIL
E260	Roy	Collins	M	RETAIL

Write a query to fetch EMP_ID, FIRST_NAME, LAST_NAME, GENDER, DEPARTMENT, and EMP_RATING if the EMP_RATING is:

- less than two
- greater than four
- between two and four

Employee rating is less than two –

SQL code –

```
select emp_id, first_name, last_name, gender, dept, emp_rating from emp_record_table where emp_rating < 2;
```

Output –

49 • `select emp_id, first_name, last_name, gender, dept, emp_rating from emp_record_table where emp_rating < 2;`

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

emp_id	first_name	last_name	gender	dept	emp_rating
E057	Dorothy	Wilson	F	HEALTHCARE	1
E532	Claire	Brennan	F	AUTOMOTIVE	1
E620	Katrina	Allen	F	RETAIL	1

No object selected

Employee rating is greater than four –

SQL code –

`select emp_id, first_name, last_name, gender, dept, emp_rating from emp_record_table where emp_rating > 4;`

Output -

50 • `select emp_id, first_name, last_name, gender, dept, emp_rating from emp_record_table where emp_rating > 4;`

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

emp_id	first_name	last_name	gender	dept	emp_rating
E001	Arthur	Black	M	ALL	5
E052	Dianna	Wilson	F	HEALTHCARE	5
E083	Patrick	Voltz	M	HEALTHCARE	5
E204	Karene	Nowak	F	AUTOMOTIVE	5

No object selected

Employee rating is between 2 & 4 –

SQL code –

`select emp_id, first_name, last_name, gender, dept, emp_rating from emp_record_table where emp_rating between 2 and 4;`

Output –

The screenshot shows the SQL Developer interface. On the left, the 'Tables' schema is expanded, listing tables like data_sci_team, data_science_team, emp_record_table, employee_record_ta, proj_table, and project_table. The 'Administration' tab is selected, and 'Schemas' is chosen. Below this, it says 'No object selected'. The main window displays a SQL query:

```
51 • select emp_id, first_name, last_name, gender, dept, emp_rating from emp_record_table
52 where emp_rating between 2 and 4;
```

The query is executed, and the 'Result Grid' shows the following data:

emp_id	first_name	last_name	gender	dept	emp_rating
E005	Eric	Hoffman	M	FINANCE	3
E010	William	Butler	M	AUTOMOTIVE	2
E103	Emily	Grove	F	FINANCE	4
E245	Nian	Zhen	M	RETAIL	2
E260	Roy	Collins	M	RETAIL	3
E403	Steve	Hoffman	M	FINANCE	3
E428	Pete	Allen	M	AUTOMOTIVE	4
E478	David	Smith	M	RETAIL	4
E505	Chad	Wilson	M	HEALTHCARE	2
E583	Janet	Hale	F	RETAIL	2

Write a query to concatenate the FIRST_NAME and the LAST_NAME of employees in the Finance department from the employee table and then give the resultant column alias as NAME.

SQL code –

Select dept, concat (first_name, ' ', last_name) as full_name from emp_record_table where dept = 'healthcare';

Output –

The screenshot shows the SQL Developer interface. On the left, the 'employee' schema is expanded, listing tables like data_sci_team, data_science_team, emp_record_table, employee_record_ta, proj_table, and project_table. The 'Administration' tab is selected, and 'Schemas' is chosen. Below this, it says 'No object selected'. The main window displays a SQL query:

```
53 • select dept, concat (first_name, ' ', last_name)
54 as fullname from emp_record_table
55 where dept = 'healthcare';
```

The query is executed, and the 'Result Grid' shows the following data:

dept	fullname
HEALTHCARE	DiannaWilson
HEALTHCARE	DorothyWilson
HEALTHCARE	PatrickVoltz
HEALTHCARE	ChadWilson

Write a query to list only those employees who have someone reporting to them. Also, show the number of reporters (including the President).

SQL Code –

```
select role, manager_id, count(*) as employee_count from emp_record_table  
group by role, manager_id order by manager_id;
```

Output -

The screenshot shows a database management tool interface. On the left, a tree view displays the database structure under 'ecommercedb', with 'employee' expanded to show tables like 'data_sci_team', 'data_science_team', 'emp_record_table', 'employee_record_ta', 'proj_table', and 'project_table'. The main area shows a SQL query with line numbers 55 to 59. The query is:

```
55 where dept = 'healthcare';  
56 • select role, manager_id, count(*) as employee_count from emp_record_table  
57 group by role, manager_id order by manager_id;  
58 • select first_name, last_name, dept from emp_record_table where dept = 'healthca  
59 union
```

Below the query, a 'Result Grid' shows the output. The grid has columns for 'role', 'manager_id', and 'employee_count'. The data is as follows:

role	manager_id	employee_count
PRESIDENT	NULL	1
MANAGER	E001	5
SENIOR DATA SCIENTIST	E083	2
ASSOCIATE DATA SCIENTIST	E083	1
LEAD DATA SCIENTIST	E103	1
ASSOCIATE DATA SCIENTIST	E103	1
LEAD DATA SCIENTIST	E428	1
SENIOR DATA SCIENTIST	E428	1
ASSOCIATE DATA SCIENTIST	E428	1
SENIOR DATA SCIENTIST	E583	2

The interface also includes a 'Filter Rows' field, an 'Export' button, and a 'Wrap Cell Content' option. The bottom of the window shows 'Result 17' and an 'Output' section with 'Action Output'.

Write a query to list down all the employees from the healthcare and finance departments using union. Take data from the employee record table.

SQL Code –

```
select first_name, last_name, dept from emp_record_table where dept = 'healthcare'  
union  
select first_name, last_name, dept from emp_record_table where dept = 'finance';
```

Output –

ecommercedb
employee
Tables
data_sci_team
data_science_team
emp_record_table
employee_record_ta
proj_table
project_table
Administration Schemas
Information

```

58 • select first_name, last_name, dept from emp_record_table where dept = 'healthcare'
59 union
60 select first_name, last_name, dept from emp_record_table where dept = 'finance';
61 • select emp_id, first_name, last_name, role, dept, emp_rating as max_rating from emp_record_table

```

first_name	last_name	dept
Dianna	Wilson	HEALTHCARE
Dorothy	Wilson	HEALTHCARE
Patrick	Voltz	HEALTHCARE
Chad	Wilson	HEALTHCARE
Eric	Hoffman	FINANCE
Emily	Grove	FINANCE
Steve	Hoffman	FINANCE

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Write a query to list down employee details such as EMP_ID, FIRST_NAME, LAST_NAME, ROLE, DEPARTMENT, and EMP_RATING grouped by dept. Also include the respective employee rating along with the max emp rating for the department.

SQL Code –

```

select emp_id, first_name, last_name, role, dept, emp_rating as max_rating from emp_record_table
where (dept, emp_rating) in (select dept, max(emp_rating) from emp_record_table group by dept) order
by dept asc;

```

Output –

Tables
data_sci_team
data_science_team
emp_record_table
employee_record_ta
proj_table
project_table
Administration Schemas
Information

```

59 union
60 select first_name, last_name, dept from emp_record_table where dept = 'finance';
61 • select emp_id, first_name, last_name, role, dept, emp_rating as max_rating from emp_record_table
62 where (dept, emp_rating) in (select dept, max(emp_rating) from emp_record_table group by dept) order by dept asc;

```

emp_id	first_name	last_name	role	dept	max_rating
E001	Arthur	Black	PRESIDENT	ALL	5
E204	Karene	Nowak	SENIOR DATA SCIENTIST	AUTOMOTIVE	5
E103	Emily	Grove	MANAGER	FINANCE	4
E052	Dianna	Wilson	SENIOR DATA SCIENTIST	HEALTHCARE	5
E083	Patrick	Voltz	MANAGER	HEALTHCARE	5
E478	David	Smith	ASSOCIATE DATA SCIENTIST	RETAIL	4
E612	Tracy	Norris	MANAGER	RETAIL	4

emp_record_table 19 x

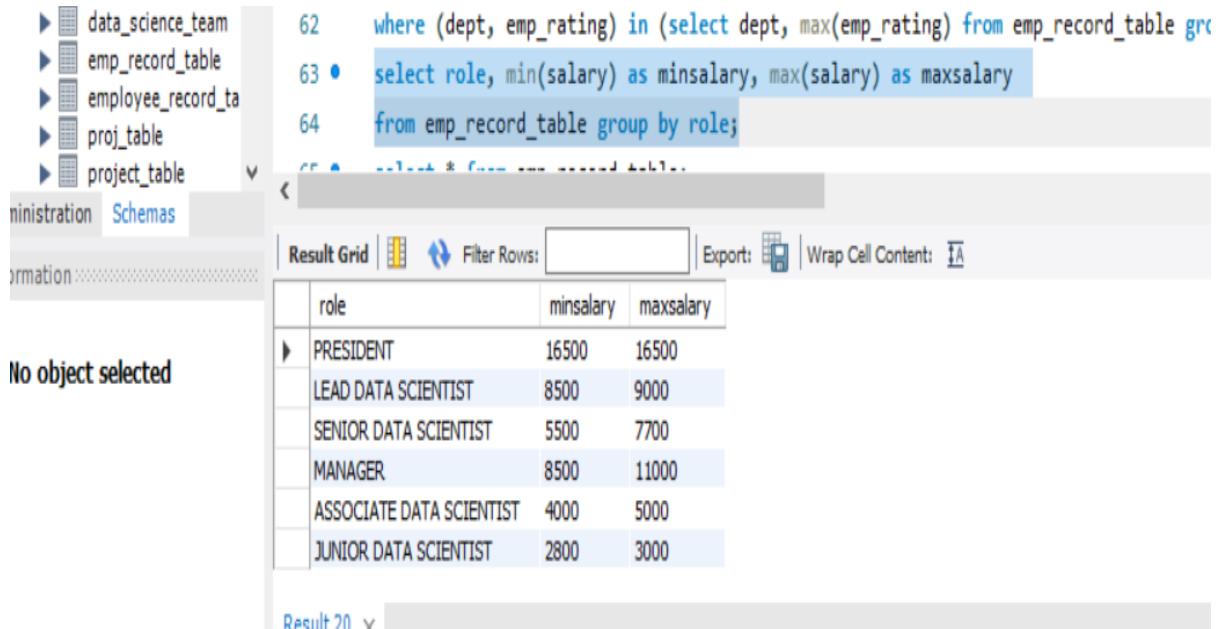
Output

- Write a query to calculate the minimum and the maximum salary of the employees in each role. Take data from the employee record table.

SQL Code –

```
select role, min(salary) as minsalary, max(salary) as maxsalary
from emp_record_table group by role;
```

Output –



The screenshot shows a database management tool interface. On the left, a tree view lists several tables: data_science_team, emp_record_table, employee_record_ta, proj_table, and project_table. The 'Schemas' tab is selected. The main area displays SQL code with line numbers 62 through 65. The code is as follows:

```
62 where (dept, emp_rating) in (select dept, max(emp_rating) from emp_record_table gr
63 • select role, min(salary) as minsalary, max(salary) as maxsalary
64 from emp_record_table group by role;
65 • select * from emp_record_table;
```

Below the code editor, there is a 'Result Grid' section. It includes a 'Filter Rows' input field, an 'Export' button, and a 'Wrap Cell Content' checkbox. The result grid displays the following data:

role	minsalary	maxsalary
PRESIDENT	16500	16500
LEAD DATA SCIENTIST	8500	9000
SENIOR DATA SCIENTIST	5500	7700
MANAGER	8500	11000
ASSOCIATE DATA SCIENTIST	4000	5000
JUNIOR DATA SCIENTIST	2800	3000

At the bottom left of the result grid, it says 'Result 70'.

- Write a query to assign ranks to each employee based on their experience. Take data from the employee record table.

SQL Code –

```
select first_name, last_name, exp as experience,
dense_rank () over (order by exp desc) exp_rank from emp_record_table;
```

Output –

Administration Schemas

data_science_team
emp_record_table
employee_record_table
proj_table
project_table

No object selected

```

65 • select * from emp_record_table;
66 • select first_name, last_name, exp as experience,
67    dense_rank () over (order by exp desc) exp_rank from emp_record_table;
68

```

Result Grid

first_name	last_name	experience	exp_rank
Arthur	Black	20	1
Patrick	Voltz	15	2
Emily	Grove	14	3
Pete	Allen	14	3
Janet	Hale	14	3
Tracy	Norris	13	4
William	Butler	12	5

Result 21 x

- Write a query to create a view that displays employees in various countries whose salary is more than six thousand. Take data from the employee record table.

SQL Code –

select first_name, last_name, country, salary from emp_record_table where salary > 6000;

Output –

Administration Schemas

data_science_team
emp_record_table
employee_record_table
proj_table
project_table

No object selected

```

68
69 • select first_name, last_name, country, salary from emp_record_table where salary > 6000;
70 • select emp_id, first_name, last_name, exp as emp_exp from emp_record_table where exp > 10 orde
71

```

Result Grid

first_name	last_name	country	salary
Arthur	Black	USA	16500
Eric	Hoffman	USA	8500
William	Butler	FRANCE	9000
Dorothy	Wilson	USA	7700
Patrick	Voltz	USA	9500
Emily	Grove	CANADA	10500
Karene	Nowak	GERMANY	7500

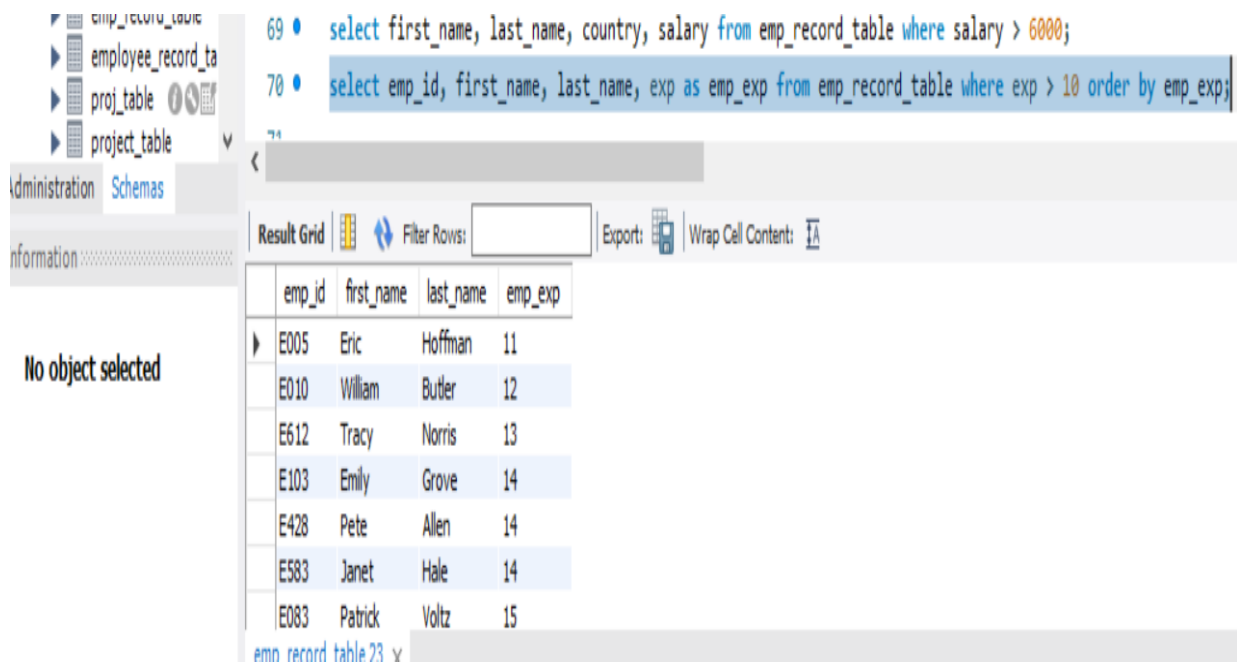
emp_record_table 22 x

- Write a nested query to find employees with experience of more than ten years. Take data from the employee record table.

SQL Code –

```
select emp_id, first_name, last_name, exp as emp_exp from emp_record_table
where exp > 10 order by emp_exp;
```

Output –



The screenshot shows a database management tool interface. On the left, a tree view displays the database schema with tables: emp_record_table, employee_record_table, proj_table, and project_table. The main area shows two SQL queries entered in a text editor:

```
69 • select first_name, last_name, country, salary from emp_record_table where salary > 6000;
70 • select emp_id, first_name, last_name, exp as emp_exp from emp_record_table where exp > 10 order by emp_exp;
```

Below the queries, a "Result Grid" displays the output of the second query. The grid has columns: emp_id, first_name, last_name, and emp_exp. The data is as follows:

emp_id	first_name	last_name	emp_exp
E005	Eric	Hoffman	11
E010	William	Butler	12
E612	Tracy	Norris	13
E103	Emily	Grove	14
E428	Pete	Allen	14
E583	Janet	Hale	14
E083	Patrick	Voltz	15

At the bottom left, a status bar indicates "No object selected".

- Write a query to calculate the average salary distribution based on the continent and country. Take data from the employee record table.

SQL Code –

```
select continent, avg(salary) as avg_salary
from emp_record_table group by continent order by continent asc;
```

Output –

The screenshot shows the SQL Developer interface. On the left, the 'Tables' folder is expanded, listing: data_sci_team, data_science_team, emp_record_table, employee_record_ta, proj_table, and project_table. The 'Administration' tab is selected, and the 'Schemas' section is active. Below this, it says 'No object selected'. The main area displays a SQL query in a text editor:

```

72 • select continent, avg(salary) as avg_salary
73 from emp_record_table group by continent order by continent asc;
74
75
76

```

Below the query editor is the 'Result Grid' tab, which shows the following data:

continent	avg_salary
ASIA	6250.0000
EUROPE	7950.0000
NORTH AMERICA	8525.0000
SOUTH AMERICA	5600.0000

At the top of the result grid, there are controls for 'Filter Rows:', 'Export:', and 'Wrap Cell Content:'.

SQL Code –

Select country, avg(salary) as avg_salary from emp_record_table
group by country order by country asc;

Output –

The screenshot shows the SQL Developer interface. On the left, the 'Tables' folder is expanded, listing: emp_record_table, employee_record_ta, proj_table, and project_table. The 'Administration' tab is selected, and the 'Schemas' section is active. Below this, it says 'No object selected'. The main area displays a SQL query in a text editor:

```

74 • Select country, avg(salary) as avg_salary from emp_record_table
75 group by country order by country asc;
76

```

Below the query editor is the 'Result Grid' tab, which shows the following data:

country	avg_salary
CANADA	7000.0000
CHINA	6500.0000
COLOMBIA	5600.0000
FRANCE	9000.0000
GERMANY	7600.0000
INDIA	6166.6667
USA	9440.0000

At the top of the result grid, there are controls for 'Filter Rows:', 'Export:', and 'Wrap Cell Content:'.

Below the result grid, there is a tab labeled 'Result 26' and an 'Output' section with an 'Action Output' dropdown.